

Aditya Kulkarni

adityakulkarnius@gmail.com | (281) 876-7066 | linkedin.com/in/aditya-kulkarni-355b81217 | github.com/Aditya-k24 | kulkarniaditya.com | Raleigh, NC

SUMMARY

ML/backend engineer with hands-on experience building high-throughput LLM inference pipelines, distributed data systems, and AI infrastructure. Scaled async LLM workloads to 2,000 req/sec with zero message loss, built multi-model routing across 8+ foundation models, and designed ML evaluation harnesses for production-grade quality measurement. Excited to work at the intersection of AI infrastructure and direct customer impact.

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Built multi-model LLM routing over 8+ models (Claude, GPT-4o, Gemini) via MCP with per-request latency/cost tracking; implemented Temporal.io durable orchestration for document processing pipelines; 394 commits, 7 production deployments.
 - Delivered TypeScript/Node.js/PostgreSQL/Redis backend features including Auth0 RBAC, real-time push (Pusher), and Sentry observability across 10+ shipped features.
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Built cloud-native genomic data platform on Google Cloud Run (500 GB, 100+ researchers); improved ML pipeline accuracy 12% via systematic hyperparameter tuning.
- Software Developer Intern

Mphasis (HP)

Jun - Aug 2024

 - Delivered health-claim ML tool (+25% accuracy via OCR/NLP); automated AWS/GCP CI/CD pipelines and added Grafana monitoring.
- AI/ML Intern

Capgemini

Jun - Aug 2023

 - Engineered Hadoop+Spark batch pipelines and Kafka real-time ingestion; cut processing time 15% and end-to-end latency 20%.

PROJECTS

- AlgoPulse

LLM Inference Pipeline

Re-architected from polling to event-driven streaming; Temporal.io durable workflow state + Kafka partitioned workers achieving 2,000 async LLM inference req/sec with zero message loss.

TypeScript · Temporal.io · Kafka · Supabase Edge Functions
- LLM Eval Harness

ML Evaluation

Vendor-neutral benchmark across Claude, GPT-4o, Gemini on 2,000 frozen examples. Tracked hallucination rate, TokenF1, p50/p95 latency, cost/call with paired bootstrap CIs, McNemar test, Holm correction. DuckDB + Parquet + Streamlit.

Python · DuckDB · Parquet · Streamlit · PyTorch
- VectorLift

Neural IR / Inference Serving

BM25 + dense bi-encoder + cross-encoder re-ranker on MS MARCO; Kafka-driven indexing into Elasticsearch + Qdrant; FastAPI serving; NDCG@10/MRR@10 evals; Prometheus + Grafana.

Python · Elasticsearch · Qdrant · Kafka · FastAPI · Prometheus
- PulseRank

Distributed Systems

CQRS event-driven leaderboard: MySQL outbox + idempotency keys, MongoDB TTL projections, Redis sorted sets (sub-ms reads); k6 benchmarks (10K users); OpenTelemetry + Prometheus + Grafana.

Node.js · MySQL · MongoDB · Redis · OpenTelemetry · k6

EDUCATION

- MS Computer Science — NC State University

Aug 2024 - May 2026

AI, OS, Data Science, Software Engineering, Algorithms
- BTech Computer Science — Mumbai University

Aug 2020 - May 2024

CGPA 9.32 / 10

SKILLS & AWARDS

Languages: Python, TypeScript, Java, SQL **ML/AI:** LLMs, RAG, LLM Evals, PyTorch, Hugging Face, Kafka, Spark, Hadoop

Infra: Docker, Kubernetes, AWS, GCP, Temporal.io, Redis, Elasticsearch, Qdrant, Prometheus, Grafana, OpenTelemetry

HackNC24 Top 10 / 100+ teams | **IEEE Published** (Mapify) | **Sanjeevani Top 1%** / 1,100+ submissions | **Google DSC Lead** (500+ students)